

Operationalisation of the choice space

Constraints

The study will be with my cohort (n=16?), and I can likely only get 10-15 minutes from each participant. I don't think I could gather enough data to create a full 10x10 map, so I'm opting to test somewhere around 6-8 decoy positions, likely where the generated model's influence maps disagree the most, chosen before any data is collected. Also, I will include a control group without any decoy present, to set the reference / baseline for decoy influence.

Operationalisation is hard. Dumbalaska et. al (2020) used rental price and property quality, after first measuring quality, instead of assuming it. When choosing what to operationalise, my original choice was 'quality', but it doesn't naturally scale because quality is subjective. Often, subscriptions like YouTube have different tiers (free, Music Premium, YouTube Premium) that scale by adding more features, and participants may feel indifferent when the service isn't real. Either I need to find a pre-existing way of scaling, or make my own. If I construct it poorly, it will make all of my results meaningless.

Priors exist. People have a general idea of how much a Netflix subscription, Amazon Prime, or iCloud storage costs. This could potentially contaminate the exact thing we're trying to measure: Context dependence.

What I'm building off of in the Dumbalska Paper

I particularly liked the two-stage process used: The first stage that maps the decision space/value of an option relative to each participant, then the second that uses three options, with the two targets being close in value.

This ensures that results are interpretable, because we are confirming that two targets have similar value for each participant first.

So, how will we operationalise?

There are a few options I've considered here:

Price x Rating

This would turn quality into something scalable. A 5-star rating is always higher than a three-star rating, and a three-star rating is always higher than a one-star rating. We see this in hotels and in online shopping products. The issue is that the rating may not mean anything really to the participants, and I would need to outline what 5-stars means compared to 3-stars, which reintroduces subjectivity.

Price x Quantitative Feature

Similar to rating, I would find a quantitative feature that can scale objectively. This could be something like GB of data in a mobile plan, games available on a game pass, or I could create my own fictional online product.

The issue with this is that utility will be non-linear. For example, A nicer house is almost always better, but participants may be indifferent to having 500GB of data every month, or 450GB, because they wouldn't get through it anyway. As of now, this would be my preferred method.

Price x Interruptions

My final idea for operationalisation is price x interruptions. I myself use Spotify Premium because I want to avoid the ads. We could tell participants that they are forced to use this service, and that as you pay more for the service, the number of interruptions will go down. A hole I see in this is that quite often interruptions are gradually frustrating. You can't look at the amount of interruptions there are to your music or video and decide how much it annoys you.

Take a product with existing data

This is my chosen method. Rather than operationalising something with arbitrary value, I've decided to find an online product with existing data, which already has value in people's minds. I would no longer have to calibrate preference either.

The Plan:

Step 1:

Find UK Sim data online. I want to find the average price/GB. This will be my single dimension scalar. Take a large amount of data and turn it into a CSV file.

Step 2:

Find the subjective amount students are willing to pay across each plan, to establish the subjective value.

Step 3:

Fit the models to classmate data, then use them on real-life pricing plans to answer: Where do these menus push you? Does these menus have decoys?